ASSIGNMENT 3

Database Design Analysis

HOSPITAL MANAGEMENT SYSTEM

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Course: BCA (Hons)

Subject: Database Management System

1. Objective

Analyze the Hospital Management System relational design for redundancy, functional dependencies, insertion/deletion/update anomalies, and improvements. The analysis uses the eight-table schema from Assignment 2: Patients, Departments, Doctors, Appointments, Medicines, Prescriptions, Bills, and Feedbacks.

2. Functional dependencies

Relation	Main functional dependency	Reason
Patients	patient_id → name, gender, dob, phone, address	A patient identifier determines that patient's profile.
Departments	department_id → department_name, location; department_name → department_id	Department ID is the primary key; department name is also unique.
Doctors	doctor_id → department_id, name, specialization, phone, email; email → doctor_id	Each doctor has a unique ID and unique email.
Appointments	appointment_id → patient_id, doctor_id, appointment_date, appointment_time, status	One appointment ID identifies one booking.
Medicines	medicine_id → medicine_name, manufacturer, price	Medicine ID identifies its catalog details.
Prescriptions	prescription_id → appointment_id, medicine_id, dosage, duration, quantity	Each prescription line has its own ID; (appointment_id, medicine_id) is also unique.
Bills	bill_id → appointment_id, bill_date, total_amount, payment_status	Bill ID identifies one bill.
Feedbacks	feedback_id → patient_id, doctor_id, rating, comments	Feedback ID identifies one feedback entry.

3. Redundancy and anomalies

A poorly designed single table might repeat patient, doctor, department, medicine, appointment and bill information on every row. The current design separates those facts into entities and connects them through keys, substantially reducing repeated storage.

Insertion anomaly

In a single combined table, a new department or medicine could not be recorded until a patient appointment existed. In the normalized design, a department can be inserted into Departments and a medicine into Medicines independently.

Update anomaly

If a doctor's phone number were repeated on every appointment row, changing it would require multiple updates and could leave conflicting values. Storing the phone once in Doctors means it is updated in one place.

Deletion anomaly

If the only appointment row containing a doctor or medicine were deleted from a combined table, the doctor or medicine details could be lost too. Separate Doctors and Medicines tables preserve master data when an appointment or prescription line is removed.

4. Normalization review

First Normal Form (1NF): each column holds a single value; multiple medicines are represented by multiple Prescriptions rows rather than a comma-separated list. **Second Normal Form (2NF):** Prescriptions uses a single-column primary key, and its dosage, duration and quantity describe that prescription line; the unique appointment/medicine pair prevents duplicate lines for the same medicine within one appointment. **Third Normal Form (3NF):** department details are stored in Departments rather than repeated in Doctors, and medicine manufacturer/price are stored in Medicines rather than repeated in each prescription.

5. Recommended improvements

- Keep all foreign keys and uniqueness constraints enabled; use CHECK constraints for ratings, prices, quantities andstatus values.
- Consider adding indexes on foreign-key columns and common search fields such as appointment_date and patient_id.
- If a real deployment needs payment installments, add a separate Payments table rather than storing multiple paymentevents in Bills. This is an extension, not part of the finalized eight-table schema.
- Define a hospital policy for deleting historical records; RESTRICT is safer than cascading deletion for medical andbilling history.
- Use transactions when creating an appointment, prescription and bill together so partial failures do not leaveinconsistent records.

Conclusion

The eight-table relational design separates independent facts, resolves the Appointment–Medicine many-to-many relationship through Prescriptions, and uses primary and foreign keys to reduce anomalies. The remaining risks are mainly operational and can be controlled through constraints, indexing, transactions and a clear retention policy.